

Tuza's conjecture for graphs of maximum degree at most eight

Yang Xu

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Abstract. We prove that every graph of maximum degree at most eight satisfies Tuza's triangle packing-covering conjecture. The argument extends Gupta's recent maximum-degree-seven theorem. It combines Puleo's reducible-set framework with an exhaustive census of eight-vertex weak König-Egervary obstructions and an exact classification of local graphs around an edge incident with a degree-eight vertex. The finite part consists of 37,172 explicit local packing-cover certificates, 208 template certificates, and two exceptional configurations that are eliminated by redirecting the selected edge. All positive certificates have passed standalone optimizer-free verification.

1. Introduction

For a finite simple graph G , let $\nu(G)$ be the maximum number of pairwise edge-disjoint triangles and let $\tau(G)$ be the minimum number of edges meeting every triangle. Tuza conjectured that $\tau(G)$ is at most twice $\nu(G)$ for every graph. Puleo proved the conjecture for graphs of maximum average degree less than seven. Gupta recently resolved the equality boundary by proving it for graphs of maximum degree at most seven [1].

The present note advances the maximum-degree threshold by one. The conceptual step is to work at a degree-eight vertex but allow the other endpoint of the selected edge to have any degree from five through eight. This forces an asymmetric local classification that is larger than the degree-seven catalogue but remains finite and exactly checkable.

Theorem 1. If $\Delta(G) \leq 8$, then $\tau(G) \leq 2\nu(G)$.

2. Reducibility and minimal counterexamples

We use Puleo's terminology [2]. A nonempty vertex set V_0 is *reducible* if there are a set S of edge-disjoint triangles and an edge set X such that (i) $|X| \leq 2|S|$; (ii) $G-X$ has no triangle meeting V_0 ; and (iii) X contains every edge of a triangle in S whose endpoints both lie outside V_0 . Puleo's reduction lemma states that if the graph remaining after the reduction satisfies Tuza's inequality, then so does G .

A graph H is weak König-Egervary (WKE) if it has a matching M and a vertex set Q , with $|Q| \leq |M|$, such that Q covers every edge of $H-M$. If the link $G[N(v)]$ is WKE, then $\{v\}$ is reducible. A graph is robust when every component of every link has at least five vertices; a minimal counterexample is robust [2].

Assume Theorem 1 is false and choose a counterexample G minimal under proper-subgraph inclusion. We may take G connected. Every proper subgraph satisfies Tuza's inequality, so G has no reducible vertex or edge set and is robust. Gupta's theorem excludes maximum degree at most seven, hence G has a vertex u of degree eight. Its link $H_u = G[N(u)]$ is connected and non-WKE.

3. The eight-vertex link census

The program `nauty-geng` generates 11,117 connected unlabelled graphs on eight vertices. Direct enumeration of all matchings M and all sets Q with $|Q| \leq |M|$ finds exactly 443 connected non-WKE graphs. Restricting to maximum degree at most four gives 1,929 graphs, exactly eight of which are non-WKE; all eight have maximum degree four.

Finite Lemma 2. Every connected non-WKE graph on eight vertices has a vertex of degree at least four.

Choose v with $\deg_{H_u}(v) \geq 4$ and put $c = |N(u) \cap N(v)| = \deg_{H_u}(v)$.

Then $4 \leq c \leq 7$ and $5 \leq \deg(v) \leq 8$. Robustness makes the link of v connected, and irreducibility makes it non-WKE.

4. The two-hub local graph

Let $C=N(u) \cap N(v)$, $A=N(u)-C-\{v\}$, and $B=N(v)-C-\{u\}$. Thus

$$|C| = c, \quad |A| = 7-c, \quad 0 \leq |B| \leq 7-c.$$

Define L on the disjoint parts $\{u,v\}$, C , A , and B by retaining all edges except possible A - B edges. Every triangle of G containing u or v is a triangle of L , and conversely every triangle of L is a triangle of G . Hence a packing-cover witness in L satisfying the three reducibility conditions makes $\{u,v\}$ reducible in the ambient graph.

For every x in C , the maximum-degree bound imposes the exact necessary budget

$$\deg_{G[C]}(x) + |N(x) \cap A| + |N(x) \cap B| \leq 6.$$

5. A generalized core template

Let R be a subset of the edges of the core $G[C]$. Let $q(R)$ be the minimum size of a vertex cover of $G[C]-R$. Let $p(R)$ be the maximum size of a two-hub edge-disjoint triangle packing supported by R : triangles may have form uvx , uxy , or vxy , with rim edge xy in R in the latter two cases.

Lemma 3 (template). If some R satisfies

$$I + |A| + |B| + |R| + 2q(R) \leq 2p(R),$$

then $\{u,v\}$ is reducible.

Proof. Choose a minimum cover Q of $G[C]-R$ and a packing S of size $p(R)$. Let X contain uv , every spoke from u to A , every spoke from v to B , every edge of R , and both hub spokes to every vertex of Q . Its size is the left side of the displayed inequality. The forced spokes cover all hub triangles using exclusive vertices; R covers the selected core edges; and Q covers the remaining core edges. Every boundary edge used by S lies in R . These are precisely the three reducibility conditions.

6. Exact local classification

For cores not handled by Lemma 3, we enumerate all connected non-WKE endpoint links of orders five through eight. We distinguish the opposite hub, retain all compatible pairs of sides satisfying the degree budget, and quotient by simultaneous core permutations, independent permutations of exclusive vertices, and hub exchange when the endpoint degrees agree.

Finite Lemma 4. Under the hypotheses of Section 4, the pair $\{u,v\}$ is reducible except for two local graphs. In both exceptions, $c=4$, $|A|=|B|=3$, and $G[C]$ is edgeless. Each exclusive side induces a triangle, and its three vertices have the same two neighbours in C . Up to isomorphism, the two cases occur when the selected two-subsets of C have intersection two or one.

After template removal, the number of unrestricted solver orbits is as follows.

c	A	B : solver orbits
4	3	0: 141; 1: 144; 2: 10,417; 3: 7,991
5	2	0: 24; 1: 3,652; 2: 11,964
6	1	0: 56; 1: 2,783
7	0	0: 2

The total is 37,174. Exact binary optimization finds certificates for 37,172 orbits. Each saved certificate records an edge-disjoint triangle set S and cover X ; a standalone verifier reconstructs the graph and checks all reducibility conditions using only set operations. Precisely two records remain. An independent Z3 encoding proves that neither has a two-hub certificate, confirming that they are genuine exceptions to this local reduction rather than solver failures.

7. Eliminating the two exceptions

In either exceptional local graph, choose a specified vertex a in A . The common neighbourhood of the edge ua consists of two vertices of C and the other two vertices of A . Its induced graph is $K_4 - e$: the two old core vertices are nonadjacent, the two exclusive vertices are adjacent, and every cross edge is present. In particular, the codegree of ua is exactly four.

Apply Finite Lemma 4 again to ua . The endpoint u still has degree eight, while $5 \leq \deg(a) \leq 8$. Both links are connected and non-WKE. The new pair cannot be exceptional, because its common-neighbour core is $K_4 - e$, whereas both exceptional cores are edgeless. Therefore $\{u, a\}$ is reducible.

No unrecorded neighbour can change this redirected core: every common neighbour of u and a lies in the completely listed degree-eight neighbourhood of u . Possible A - B edges and edges leaving the local vertex set are therefore irrelevant.

8. Proof of Theorem 1

Let G be the minimal counterexample from Section 2. Select a degree-eight vertex u . Finite Lemma 2 supplies an incident edge uv of codegree between four and seven. The endpoint links satisfy the hypotheses of Finite Lemma 4. If the corresponding local graph is not exceptional, $\{u, v\}$ is reducible. If it is exceptional, Section 7 supplies a reducible pair $\{u, a\}$. Either conclusion contradicts the absence of reducible sets in a minimal counterexample. Hence no counterexample exists.

9. Verification record

The computation used Python 3.13.2, NumPy 2.3.1, SciPy 1.16.2, Z3 5.1.0, and nauty 2.9.0. The 37,174-record orbit catalogue was independently regenerated and matched the original byte for byte, with SHA-256 74C38803093DA7BCEDB43A728E32D6AF73CF3314119E187160A093708A21D43A.

The optimizer-free verifier accepted all 37,172 unrestricted positive certificates and reported only indices 10,708 and 10,709. A second optimizer-free verifier accepted 208 template certificates. The two exceptions and their redirected $K_4 - e$ cores were checked independently with Z3. The full eight-vertex pass reproduced 11,117 connected unlabelled graphs and 443 connected non-WKE links.

The source programs, JSON certificates, checksums, environment details, and exact reproduction commands accompany this draft. Computational verification is complete; independent reproduction and peer review remain necessary before the result should be treated as established literature. Computational exploration and manuscript preparation were assisted by OpenAI Codex; the released exact certificates and standalone verifiers are provided to make every finite claim independently checkable.

References

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